Appendix A

PhilFraud-SS Annotation Manual

Screenshot-Based Philippine Digital Fraud Dataset

Version 1.0 | De La Salle University — Master's Thesis

PhilFraud-Agent: Adaptive Multimodal LLM Agentic System for Philippine Digital Fraud Detection

How to use this manual

- Read Sections 1–3 before you begin annotating.
- Keep Sections 4–5 open during your session as a reference.
- Section 6 covers quality control procedures managed by the lead researcher. Section 7 contains the edge case encyclopedia — consult it whenever you are unsure.

A.1 Purpose and Overview

Why Your Labels Matter

PhilFraud-SS is the ground truth dataset that will be used to train, calibrate, and evaluate the PhilFraud-Agent system, an agentic LLM built to detect evolving digital fraud in the Philippine digital ecosystem. Every label you assign directly determines what the system learns to distinguish as fraud versus legitimate content, and how it reports confidence in its decisions.

Your annotations will serve three concrete functions:

1. **Ground truth for model evaluation** — your Binary Label determines whether the system's fraud/safe prediction is correct or incorrect.
2. **Episodic memory seeding** — confirmed fraud cases from the training split are stored as templates the agent retrieves during inference.
3. **Rationale quality benchmarking** — your written rationale becomes the reference against which the LLM-as-a-Judge rubric is validated.

Who Should Annotate

Annotators must satisfy at least one of the following qualifications:

- Background in cybersecurity, information security, or digital forensics
- Experience in consumer protection, law enforcement, or Philippine cybercrime investigation
- Active Filipino social media user with documented exposure to online fraud reporting (e.g., moderator of a scam-reporting group, or prior work involving online fraud cases)

You do not need to be a machine learning expert. You are being recruited for your domain knowledge and judgment, not technical AI skills.

A.2 Annotation Task Summary

For each screenshot, you will assign **five items** (shown in Table A.1):

Estimated time: Easy cases: ˜1 minute. Medium cases: ˜2 minutes. Hard cases: ˜3–5 minutes.

Session recommendation: Do not annotate more than 100 cases per session without a break. Annotation fatigue degrades label quality.

Table A.1: Annotation Fields and Options

#	Field	Type	Options
1	Binary Label	Choice	Fraud / Safe
2	Fraud Category	Choice	Phishing / Smishing / Impersonation / Loan or Investment / Legitimate
3	Difficulty Tier	Choice	Easy / Medium / Hard
4	Rationale	Free text	1–2 sentences
5	Confidence	Choice	High / Medium / Low

A.3 Label Definitions

A.3.1 Binary Label

Fraud — The screenshot provides evidence of a deceptive, misleading, or fraudulent attempt designed to extract money, personal data, credentials, or trust from a target.

Safe — The screenshot shows content from a legitimate source. This includes real promotional material, official bank notifications, government advisories, and legitimate product advertisements — even if the content uses urgency language or promotional offers.

The most important boundary in this dataset: Aggressive-but-legitimate marketing is **Safe**. A real GCash promotion with a countdown timer is Safe. A fake GCash promotion designed to steal credentials is Fraud. The difference lies in intent and authenticity, not in tone or urgency alone.

When in doubt, ask: Would a legitimate Philippine institution (BSP-regulated bank, DTI-registered company, SEC-registered entity) plausibly send or post this content through this channel?

A.3.2 Fraud Category

Assign a category **only if Binary Label is Fraud**. If Binary Label is Safe, select Legitimate.

Phishing / Smishing

Fraudulent messages designed to steal credentials, OTPs, or personal information. Typically sent via SMS, email, Messenger, Telegram or Viber. Common signals:

- Requests for OTP, password, or account verification
- Links to fake login pages mimicking GCash, BDO, BPI, Maya, or government portals
- Fake reward or prize claims requiring account submission
- Urgency framing: "Your account will be suspended," "Verify now or lose access"

Impersonation

Content where a person, brand, or institution is being falsely represented. Common signals include fake celebrity, influencer, or public official endorsements, fake official Facebook pages or groups mimicking known brands, scammers posing as relatives or officials, and fabricated screenshots of news articles or official statements.

Loan / Investment Scam

Fraudulent offers of financial products — loans, investments, or income opportunities — designed to extract money or recruit victims. Common signals:

- Unrealistic returns (e.g., "Double your money in 7 days")
- Fake DTI, SEC, or BSP approval or registration numbers
- Pyramid scheme or Multi-level marketing (MLM)
- Loan offers requiring upfront fees before disbursement
- Fake income screenshots or fabricated dashboards

A.3.3 Difficulty Tier

Assign the tier based solely on what this single screenshot reveals when examined independently, even if it came from a multi-image post.

Easy

The screenshot contains **three or more independent, self-evident fraud signals** that do not require significant inference or cross-referencing. A non-expert viewer would likely identify this as fraud within 15–30 seconds. Characteristics:

- Multiple compound signals present simultaneously (e.g., impersonated brand + urgency + external link + prize claim)
- Obvious visual anomalies (low-resolution logo, off-brand colors, wrong font)
- Explicit credential or OTP request
- Clearly non-official contact channel (e.g., Gmail address used for a "BDO" notification)

Medium

The screenshot contains **one or two fraud signals** that require some inference, contextual knowledge, or Taglish interpretation to identify. A careful observer would likely identify this as suspicious, but it is not immediately obvious. Characteristics:

- Single signal without full convergence (e.g., urgency language without brand impersonation)
- Requires familiarity with legitimate Philippine platform UX to spot the inconsistency
- Taglish phrasing with subtle scam markers ("Libreng load para sa mga napili")
- Promotional tone that could be either aggressive marketing or fraud

Hard

The screenshot contains **minimal, subtle, or highly sophisticated fraud signals** that could fool even careful observers. The content closely mimics legitimate material. Characteristics:

- Pixel-accurate brand impersonation (correct logo, correct colors, plausible layout)
- Edited screenshot or fabricated transaction record with plausible formatting
- Minimal text with no explicit credential request or urgency language
- Content that appears entirely legitimate in isolation
- Adversarial design that exploits platform-specific visual conventions

Multi-image post rule: If this screenshot came from a multi-image post, assign its tier based on this image alone. If it appears legitimate in isolation but is suspicious only because of the other images in the post, assign **Medium or Hard** (not Easy) and note this in your Rationale.

A.3.4 Rationale

Write **1–2 sentences** describing the main signal(s) that led to your Binary Label decision. Be specific.

Good rationale examples:

> "The screenshot displays a GCash-branded notification requesting OTP verification via a Gmail sender address, and includes a countdown timer with an external shortened URL — three independent signals converging on phishing."

> "The investment offer claims SEC registration but the certificate number format does not match the official SEC format, and the promised 30% monthly return is implausible for any regulated financial product."

> "This appears to be a legitimate BPI promotional push notification. The sender domain matches official BPI channels, no credential input is requested, and the offer contains a valid terms and conditions reference."

What to avoid:

- Vague rationales: "Looks like a scam" or "Seems legit"
- Repeating the label without explaining evidence: "This is fraud because it is fraudulent"
- Copying the fraud category name only: "This is phishing"

A.3.5 Confidence

Rate your own certainty in your Binary Label decision.

Table A.2: Confidence Level Interpretation

Level	Meaning
High	You are certain. The evidence is clear and you are not second-guessing your label.
Medium	You are reasonably sure, but one or two signals are ambiguous or you could imagine an argument for the other label.
Low	You are genuinely uncertain. The case is borderline and you would not be surprised if the adjudication goes the other way.

A.4 Step-by-Step Annotation Procedure

Follow this procedure for every case without exception.

1. **View the full image.** Zoom in to read all text clearly. Look at the sender, the URL or contact information, the visual layout, any logos, and the nature of the request or offer.

2. **Assign Binary Label.** Ask yourself: Is this attempting to deceive me into surrendering credentials, money, or personal information? Or is this content from a legitimate source?

3. **Assign Fraud Category.** If Fraud: identify the primary deception mechanism (credential theft → Phishing/Smishing; false identity → Impersonation; financial exploitation → Loan/Investment). If Safe: select Legitimate.

4. **Assign Difficulty Tier.** Count independent fraud signals. 3+ clear signals = Easy. 1–2 signals requiring inference = Medium. Minimal or adversarially sophisticated signals = Hard. If from a multi-image post and ambiguous in isolation, default to Medium or Hard.

5. **Write your Rationale.** Identify the one or two most important specific signals that drove your Binary Label. Name them explicitly (e.g., "Gmail sender domain on a BDO-branded notification" rather than "wrong email").

6. **Rate your Confidence.** Be honest. Low confidence cases are not failures — they are valuable signal about genuine case ambiguity.

7. **Submit.** Click Submit in Label Studio. Do not go back and change labels after submission unless you accidentally selected the wrong radio button.

A.5 Fraud Signal Reference Guide

Use this as a quick-reference lookup during annotation.

A.5.1 Visual Signals

Use table A.3 to quickly check visual cues in the screenshot that might indicate fraud or legitimacy.

A.5.2 Textual Signals (including Taglish)

Use table A.4 to identify suspicious or reassuring wording, including Taglish phrases, in the message content.

Table A.3: Visual Signals

Signal	What to look for
Logo impersonation	Stretched, outdated, low-res, or slightly off-color version of GCash, BDO, BPI, Maya, SSS, PhilHealth, etc.
Urgency design	Red/orange color dominance, countdown timer graphics, “LIMITED TIME” banners
QR code presence	QR codes in unexpected contexts (payment requests from unofficial channels)
Layout cloning	Screen layout that closely mimics the GCash app, BDO online banking, or official SMS format
Missing official markers	No footer, no privacy policy reference, no official contact details
Fabricated document	Fake DTI certificate, fake SEC registration, fake BIR receipt with incorrect formatting
Image manipulation artifacts	Signs of photo editing such as mismatched fonts, uneven spacing/alignment, edited timestamps or amounts, blurred regions, inconsistent lighting, cropping artifacts, or overlayed text elements
Suspicious platform/interface	Fake crypto trading dashboards, unofficial investment apps, cloned exchange websites, fabricated wallet balances, unrealistic profit displays, or interfaces mimicking legitimate platforms (e.g., Binance, Coins.ph, MetaMask)
Logo/account impersonation	Use of copied or stolen branding, profile pictures, page names, posts, or visual assets to imitate legitimate companies, government agencies, celebrities, relatives, or local officials; includes cloned Facebook pages/accounts and reposted official content presented as authentic

A.5.3 Structural Signals

Use table A.5 to assess how the message is delivered and structured, and whether that structure fits normal behavior for the claimed sender.

A.6 Quality Control Procedures

This section is primarily for the lead researcher but is included here for transparency.

Table A.4: Textual Signals

Signal	Examples
Urgency language	"Mag-claim na," "Mawawala ang iyong account," "Last chance," "Verify now," "Expires today"
Reward/prize claims	"Nanalo ka," "Libreng load," "Cash prize," specific peso amounts as winnings
Authority impersonation	Unauthorized use of BSP, BDO, GCash, SSS, PhilHealth, DICT, NBI, PNP in message text
Anomalous contact	Gmail or Yahoo address for a "bank" notification; non-official phone numbers
OTP/credential request	Any request for PIN, OTP, password, or CVV outside the official app
Investment framing	"Guaranteed returns," "Passive income," "Doble ang pera," "Pagtutulungan"
Language anomalies	Misspellings, unusual capitalization (e.g., "FrEe LoAd"), excessive punctuation, grammatical inconsistencies, awkward phrasing, or mixed Filipino-English syntax inconsistent with official communications

Table A.5: Structural Signals

Signal	What to look for
Platform mismatch	Bank branding in SMS, Messenger, or Facebook — official bank comms do not come through these
Shortened URLs	`http://bit.ly`, tinyurl, or other redirects in a "bank notification"
Non-brand domain	Links to domains that do not match the claimed brand (e.g., `gcash-reward.xyz`)
Missing verification path	No reference to a verifiable official channel (no app store link, no official website)

A.6.1 Calibration Session (Before Full Annotation)

Before distributing all approx. 396 cases, all three annotators will complete a **20-case calibration batch** on a stratified sample (5 Easy, 5 Medium, 5 Hard fraud cases, 5 Safe cases). After independent annotation:

1. Fleiss' kappa is computed on the calibration batch.
2. The lead researcher facilitates a 30–45 minute group discussion on all dis-

agreement cases.

3. Annotation guidelines are updated if the discussion reveals any ambiguity in definitions.
4. A revised version of this manual (if any updates were made) is re-distributed before full annotation begins.

Calibration kappa target: $\kappa \geq 0.60$ (substantial agreement) before full annotation proceeds. If calibration kappa is below 0.60, a second calibration batch of 10 cases is run after further guideline clarification.

A.6.2 Inter-Annotator Agreement Targets

Table A.6: Inter-Annotator Agreement Targets

Subgroup	**Minimum Acceptable κ**	**Rationale**
Overall	$\kappa \geq 0.80$	Almost perfect agreement — required for high-quality ground truth
Easy tier	$\kappa \geq 0.85$	Easy cases should yield near-unanimous agreement
Medium tier	$\kappa \geq 0.75$	Moderate ambiguity expected
Hard tier	$\kappa \geq 0.61$	Substantial agreement — genuine ambiguity accepted

Hard-tier kappa between 0.61–0.80 is **not a failure**. It reflects genuine case ambiguity, which is epistemically informative and will be reported separately in the thesis.

A.6.3 Disagreement Adjudication Protocol

Tier 1 — Automatic (2 vs. 1 disagreement): The majority label is used automatically. The frequency of Tier 1 resolutions is logged and reported as a transparency measure in the dataset card.

Tier 2 — Facilitated Discussion (3-way disagreement): A facilitated discussion session is held. The lead researcher presents the case without revealing individual annotator positions. The goal is to determine whether the disagreement stems from ambiguous annotation guidelines or genuine case ambiguity.

Tier 3 — Fourth Expert Adjudication: Cases that remain unresolved after the Tier 2 discussion are reviewed by a pre-designated fourth domain expert who was not part of the original annotation team. These cases are retained in the dataset with `contested_flag = True` and reported as a separate analysis subgroup.

A.6.4 Quality Control Checkpoints

The lead researcher will run the following checks after each batch of 100 completed cases per annotator:

Table A.7: Quality Control Checkpoints

Check	**Threshold**	**Action if failed**
Per-annotator fraud rate	Within ±15% of the other annotators' rate	Flag for individual discussion
Rationale length	Mean > 15 words per rationale	Remind annotator of rationale requirements
Confidence distribution	Not > 80% High across all cases	Remind annotator of confidence calibration purpose
Annotation speed	Not < 30 seconds average per case	Flag potential inattentive labeling

A.6.5 Label Studio Configuration Checklist

Before distributing to annotators, the lead researcher confirms:

- All 396 cases loaded with correct `case_id` and image file
- All three annotators assigned as **Annotators** (not Managers) — prevents label visibility
- Instructions popup populated with the Quick Reference (Section 6 of this manual condensed)

- “Show instructions before labeling” enabled for first-time access
- Annotation config validated: all five fields present, choices match this manual exactly
- PII redaction confirmed on all images and OCR text
- Pilot batch of 20 cases designated and loaded first

A.7 Edge Case Encyclopedia

This section documents known difficult cases and the correct annotation approach for each.

Edge Case 1: Aggressive but Legitimate Marketing

Description: A real promotional post from a Philippine company (e.g., Shopee, Lazada, GCash) uses extremely urgent language, countdown timers, and large prize claims — but the branding is authentic and no credential input is requested.

Rule: Label as **Safe**. Urgency and reward language alone do not constitute fraud. The test is whether the content is designed to deceive. Legitimate promotions that comply with DTI requirements are Safe even if they feel manipulative.

Edge Case 2: Screenshot of a Screenshot

Description: The annotated image is a screenshot of another person’s screenshot — for example, a user posted a photo of their phone screen showing a scam SMS.

Rule: Annotate based on the **content of the inner screenshot** (the scam message), not the outer framing. If the inner message contains fraud signals, label as Fraud. The meta-context (someone reporting a scam) does not change the label.

Difficulty tier: Base the tier on the signals visible in the inner screenshot, not on the quality of the outer photo.

Edge Case 3: Partial or Cropped Screenshot

Description: The image shows only part of a message, cutting off the sender, URL, or key text.

Rule: Annotate based only on what is visible. If the visible content contains sufficient signals to determine a label, assign it. If the crop removes too much context to make a reliable judgment, assign Fraud or Safe using your best judgment, Difficulty Tier = **Hard**, Confidence = **Low**, and note in Rationale that it is a partial screenshot with missing context.

Edge Case 4: Screenshots from Multi-Image Posts

Description: You know (or suspect) that this image came from a post that contained multiple screenshots. The image appears legitimate or ambiguous in isolation.

Rule: Assign difficulty based on this single image alone. If it appears legitimate in isolation but you know it was accompanied by obvious fraud evidence in the same post, assign **Medium or Hard** (never Easy) and note that it appears legitimate in isolation but context from co-posted images suggests fraud.

Edge Case 5: Taglish Content Without Visual Fraud Signals

Description: The screenshot contains only Taglish text — no branding, no logos, no URLs. The message uses scam-associated language ("Mag-claim na ng inyong premyo," "Kailangan ng iyong OTP") but there are no visual fraud signals.

Rule: Textual signals alone are sufficient for a Fraud label. Assign Fraud with Phishing/Smishing or Impersonation category depending on content, and set difficulty based on the number and clarity of textual signals.

Edge Case 6: Legitimate-Looking Investment Content

Description: A screenshot shows an investment opportunity with a polished design, professional language, and no explicit red flags — but the promised returns are unusually high.

Rule: If returns are demonstrably implausible (e.g., 30% monthly, 100% in 7 days) label as Fraud, Loan/Investment, with Medium or Hard difficulty depending on how polished the presentation is. If returns are high but plausibly within legitimate high-risk ranges and no other signals are present, label Safe with Low confidence and note ambiguity. If the entity is verifiable as SEC-registered and the return claims are within plausible ranges, label Safe.

Edge Case 7: Government Warning or Advisory Screenshot

Description: The screenshot appears to be an official government warning about scams (e.g., a BSP or NBI advisory posted to Facebook).

Rule: Label as **Safe**. Government advisories warning about fraud are legitimate content. Do not label a fraud warning as Fraud. Exception: if the "government advisory" is itself a fake — fabricated to impersonate a government agency and direct victims to a fraudulent link — label as Fraud (Impersonation).

Edge Case 8: Image Is Blank, Corrupted, or Unintelligible

Description: The image fails to load, is entirely black, is too blurry to read, or contains content completely unrelated to fraud or financial services.

Rule: Do **not** assign a label. Type `SKIP - [reason]` in the Rationale field and submit with Low confidence. The lead researcher will review and either replace the image or remove the case from the dataset.

Edge Case 9: Deepfake or AI-Generated Content

Description: A video thumbnail or static image appears to show a public figure (celebrity, government official, bank executive) endorsing an investment product or financial platform.

Rule: Label as **Fraud, Impersonation, Hard**. Fabricated celebrity or official endorsements are among the most adversarially sophisticated fraud types. Even if you cannot definitively confirm it is AI-generated, an unprecedented or implausible endorsement is sufficient for a Fraud label at High confidence.

Edge Case 10: Real Scam But Screenshot Is from a News Article

Description: The screenshot shows a scam message as displayed within a Philippine news article, with the news site branding visible around it.

Rule: The inner content is a scam example, but the context makes it legitimate educational content. Label as **Safe** — the purpose of the screenshot is to warn, not to deceive. Note in Rationale that it is scam content displayed within a legitimate news article for educational or warning purposes.

Edge Case 11: Impersonation vs. Phishing/Smishing

Description: Cases where both a fake identity (brand or person) and a possible credential-harvesting attempt appear in the same screenshot.

Rule: Use the **primary mechanism** test. If credential harvesting (visible link, OTP request, login form, or explicit instruction to enter credentials) is present, tag as **Phishing/Smishing**. If the screenshot only shows a false identity without a visible credential-harvesting mechanism, tag as Impersonation. Tiebreaker: If you removed the brand, would the fraud still work? If yes, tag Phishing/Smishing; if **no**, tag **Impersonation**.

Edge Case 12: Task scams

Description: "Task scam" workflows where victims are asked to perform simple online tasks (liking posts, joining channels, etc.) and then pressured to make deposits to unlock fake earnings.

Rule: Always tag as **Loan/Investment**. The defining mechanism is financial extraction through fake earnings and deposits, even if brand impersonation is used only for recruitment.

Edge Case 13: Impersonation vs. Phishing/Smishing

Description: Harassing SMS messages sent by Online Lending Apps (OLA) or their agents to borrowers, contacts, or uninvolved third parties, often containing threats or shaming.

Rule: Default to **Loan/Investment** — this is a post-disbursement tactic of a predatory lending operation, even when sent to uninvolved contacts or co-makers. Exception: If the SMS clearly impersonates a lawyer, NBI, PNP, or a court (e.g., "From NBI," "From Atty. X," "From RTC Branch X"), tag as **Impersonation**.

Edge Case 14: Cropped/Partial Phishing Email (No Link Visible)

Description: Screenshots of emails where the phishing layout is partly cut off and the link or button is not visible, but the body appears suspicious.

Rule: Tag as **Impersonation** if no credential-harvesting mechanism is visible in the screenshot. Exception: If the email body **explicitly references** a link or button (e.g., "Click the link below," "Click the button to verify") even if the button itself is cropped out, tag as **Phishing/Smishing** — the intent to harvest credentials is stated. Default difficulty: **Hard**.

Edge Case 15: Unconfirmed Fake Social Media Page

Description: Social media pages that strongly appear to be fake or cloned, but where the annotator cannot directly confirm the real entity being impersonated.

Rule: Apply a **2-signal threshold**: if **two or more** red flags are present (e.g., misspelled brand name, off-brand or mismatched branding, no external link to an official site/app, suspicious follower count or engagement), tag as **Impersonation** even without confirmed identity of the original entity. Meta Verified badges and high follower counts are **not** exculpatory — both can be purchased or inherited via account hijacking. Default difficulty: **Hard**.

Edge Case 16: Gambling Scams

Description: Fake or rigged gambling schemes such as fraudulent casino apps, betting platforms, "VIP tips" groups, or fixed-game channels that lure victims with initial fake wins and then drain deposits.

Rule: Tag as **Loan/Investment**. Gambling scams follow the same financial extraction pattern as investment fraud: victims are induced to deposit money they never recover. If a celebrity image or famous streamer is used to promote a fake gambling platform, apply the primary mechanism test: if the **platform** is the fraud, tag **Loan/Investment**; if the screenshot is a standalone fake celebrity endorsement with no platform shown, tag **Impersonation**.

A.8 Quick Reference Card

(Condensed for use during annotation sessions)

BINARY LABEL
Fraud → Designed to deceive and extract credentials, money, or personal data
Safe → Legitimate content, even if urgent or promotional

FRAUD CATEGORY (Fraud only)
Phishing/Smishing → Credential or OTP theft via link or direct request
Impersonation → False identity: fake brand, fake person, fake official
Loan/Investment → Fraudulent financial offer, pyramid scheme, fake SEC cert
NA → If Safe

DIFFICULTY TIER (this screenshot only)
Easy → 3+ self-evident independent signals, non-expert can spot it quickly
Medium → 1–2 signals requiring inference or Taglish knowledge
Hard → Sophisticated, adversarial, minimal signals, or ambiguous in isolation

RATIONALE (required)
Name the specific signal(s) that drove your Binary Label; 1–2 sentences, be precise

CONFIDENCE
High = certain | Medium = reasonably sure | Low = genuinely borderline

REMEMBER:
Urgency alone ≠ Fraud; assign difficulty based on this image alone; for multi-image posts, if ambiguous in isolation → Medium or Hard minimum; if unsure between Safe and Fraud → Low confidence and explain in Rationale.

A.9 Annotator Agreement Form

By beginning annotation on Label Studio, you confirm that you have:

- Read and understood Sections 1–6 of this manual
- Completed the 20-case calibration batch
- Attended the calibration debrief session
- Agreed not to discuss individual case labels with other annotators until post-annotation agreement analysis
- Understood that your labels, rationales, and confidence scores may be quoted in the thesis (without identifying which annotator provided them)

Annotator pseudonym (for thesis reporting): A1 / A2 / A3 (assigned by lead researcher).

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